DecAug: Out-of-Distribution Generalization via Decomposed Feature Representation and Semantic Augmentation

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Introduction

Out-of-distribution Generalization:

- Correlation shift
- Diversity shift

Approach: DecAug

- Decomposed feature representation
- Gradient-based semantic augmentation

Two-dimensional OoD

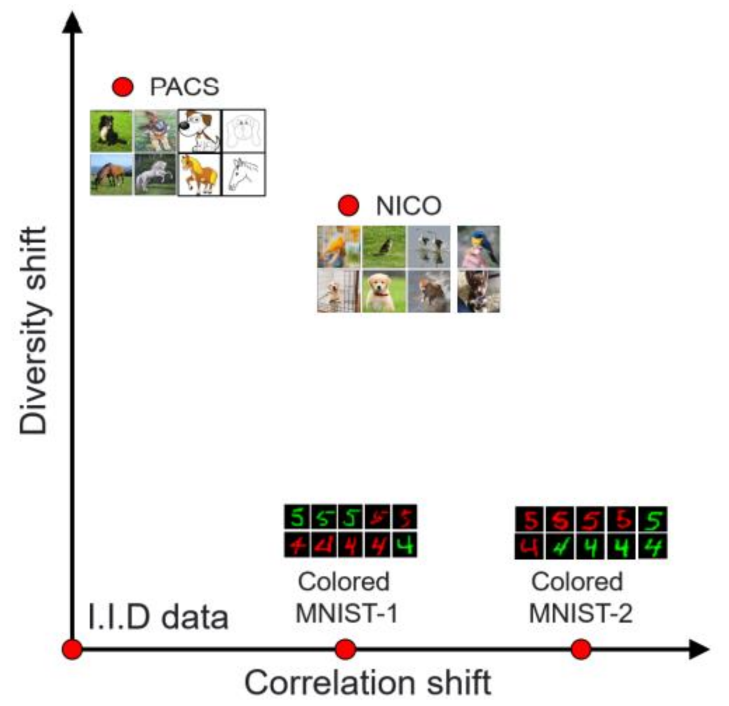
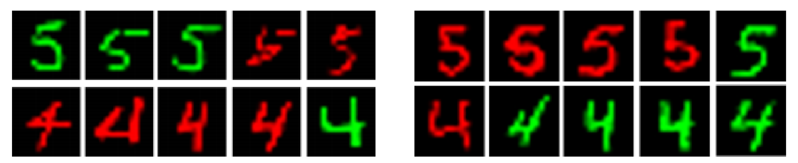


Illustration of the two-dimensional out-of-distribution problem among datasets. X-axis denotes the correlation shift which controls the correlation proportions of the correlated features, Y-axis denotes the diversity shift which stands for the change of feature types.

Deep Learning has demonstrated superior performance on standard benchmark datasets from various fields. However, How to deal with OoD generalization is still an open problem:

- **Correlation shift.** One is the correlation shift, which means that labels and the environments are correlated, and the relations change across different environments.
- **Diversity shift.** Data in different domains have significantly different styles. The model trained on the training set would susceptible to the diversity shift on the test set.
- **Two-dimension OoD.** Data in actual scenarios usually involve two different OoD factors simultaneously

Typical Images Illustration



(a) Train (b) Test

Typical examples of out-of-distribution correlation shift data from the Colored MNIST dataset.

(a) P (b) A (c) C (d) S

Typical examples of out-of-distribution diversity shift data from the PACS dataset.

(a) In cage (b) In water (c) On grass (d) Others

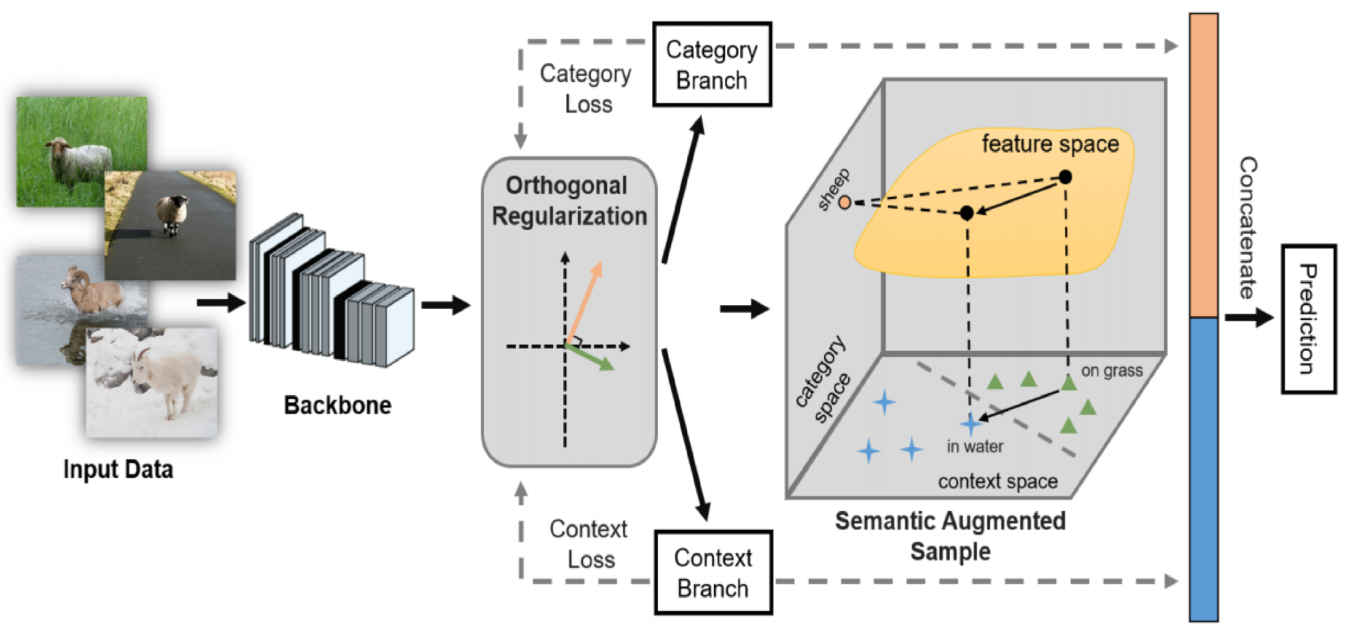
Some examples of the two-dimensional out-of-distribution data from the NICO dataset.

Related Works

Review literature related to risk regularization methods, domain generalization, stable learning, data augmentation and disentangled representation.

- **Risk regularization methods for OoD generalization.** Methods such as IRM, IRM-Games, Rex typically perform well on synthetic datasets. However, it is unknown how they can generalize on more complex practical datasets.
- **Domain generalization.** Recent OoD research finds that domain adaptation methods with similar design principles can have problems with training and test distribution is too large.
- **Stable learning.** While these methods can have theoretical guarantees on simplified models, this method may not be able to work well especially in the deep learning paradigm.
- **Data augmentation.** Compared with related DA methos, our method performs gradient-based augmentation on disentangled context-related features to eliminate distribution shifts for various OoD tasks.
- **Disentangled representation.** Disentangling the latent factors from the image variants is a promising way to provide an understanding of the observed data. In this paper, semantic vectors found by DecAug with orthogonal constrains are disentangled from each other in the feature space.

Methods



An overview of the proposed DecAug.

Algorithm 1 DecAug: Decomposed Feature Representation and Semantic Augmentation for OoD generalization

Input: Training set $\mathcal{D}$, batch size n, learning rate β, hyper-parameters $\epsilon, \lambda^1, \lambda^2, \lambda^{orth}$.
Output: $\theta, \theta^1, \phi^1, \theta^2, \phi^2, \phi$.

Initialize $\theta, \theta^1, \phi^1, \theta^2, \phi^2, \phi$;
repeat
  Sample a mini-batch of training images $\{(x_i, y_i, c_i)\}_{i=1}^n$ with batch size n;
  for each (x_i, y_i, c_i) **do**
    $z_i \leftarrow g_\theta(x_i)$;
    $\mathcal{L}_i^1(\theta, \theta^1, \phi^1) \leftarrow \ell(h_{\phi^1} \circ f_{\theta^1}(z_i), y_i)$;
    $\mathcal{L}_i^2(\theta, \theta^2, \phi^2) \leftarrow \ell(h_{\phi^2} \circ f_{\theta^2}(z_i), c_i)$;
    Compute $\mathcal{L}_i^{orth}(\theta^1, \phi^1, \theta^2, \phi^2)$ according to Eq. (1);
    Randomly sample α_i from $[0, 1]$;
    Generate $\tilde{z}_i^2$ according to Eq. (2);
    $\mathcal{L}_i^{concat}(\theta, \theta^1, \theta^2, \phi) \leftarrow \ell(h_\phi([f_{\theta^1}(z_i), \tilde{z}_i^2]), y_i)$;
    Compute $\mathcal{L}_i(\theta, \theta^1, \phi^1, \theta^2, \phi^2, \phi)$ according to Eq. (3);
  end for
  $(\theta, \theta^1, \phi^1, \theta^2, \phi^2, \phi) \leftarrow (\theta, \theta^1, \phi^1, \theta^2, \phi^2, \phi) - \beta \cdot \nabla \frac{1}{n} \sum_{i=1}^{n} \mathcal{L}_i(\theta, \theta^1, \phi^1, \theta^2, \phi^2, \phi)$;
until convergence;

To deal with the aforementioned two different types of distribution shifts simultaneously, we argue that it is critical to obtain the decomposed features., one is essential for predicting category and the other is other is not essential but correlated for recognition.

- **Feature decomposition.**
  It is known that the direction of the non-zero gradient of a function is the direction in which the function increases most quickly, the direction that is orthogonal to the gradient direction is the direction in which the function increases most slowly. To better decompose the features into category-related and context-related features, we enforce the gradient of the category loss to be orthogonal to the gradient of the context loss
- **Semantic augmentation.**
  Considering that context-related features cause correlation or diversity shifts in our setting, we perform augmentation on the context-related features to eliminate such kind of distribution shifts. To ensure good performances across different environments, we postulate a worse case for the model to learn for OoD generalization by calculating the adversarially perturbed examples in the feature space.

Results

- **Results on the PACS datasets**

Model	Art Painting	Cartoon	Sketch	Photo	Average
ERM (Carlucci et al. 2019)	77.85	74.86	67.74	95.73	79.05
IRM (Arjovsky et al. 2019)*	70.31	73.12	75.51	84.73	75.92
Rex (Krueger et al. 2020)*	76.22	73.76	66.00	95.21	77.80
JiGen (Carlucci et al. 2019)	79.42	75.25	71.35	**96.03**	80.51
DANN (Ganin et al. 2016)	81.30	73.80	74.30	94.00	80.80
MLDG (Li et al. 2017b)	79.50	77.30	71.50	94.30	80.70
CrossGrad (Shankar et al. 2018)	78.70	73.30	65.10	94.00	80.70
MASF (Dou et al. 2019)	80.29	77.17	71.69	94.99	81.03
Cumix (Mancini et al. 2020)	**82.30**	76.50	72.60	95.10	81.60
DecAug	*79.00*	***79.61***	***75.64***	*95.33*	***82.39***

* Implemented by ourselves.

- **Evaluation on Colored MNIST dataset**

Model	Acc test env
ERM*	17.10 ± 0.6
IRM (Arjovsky et al. 2019)	66.90 ± 2.5
Rex (Krueger et al. 2020)	68.70 ± 0.9
F-IRMGames (Ahuja et al. 2020)	59.91 ± 2.7
V-IRMGames (Ahuja et al. 2020)	49.06 ± 3.4
ReBias (Bahng et al. 2020)*	29.40 ± 0.3
JiGen (Carlucci et al. 2019)*	11.91 ± 0.4
DecAug	$\boldsymbol{69.60 \pm 2.0}$
ERM, grayscale model(oracle)	73.00 ± 0.4
Optimal invariant model (hypothetical)	75.00

* Implemented by ourselves.

- **Comparison on the NICO dataset**

Model	Animal	Vehicle
ERM*	75.87	74.52
IRM (Ahuja et al. 2020)*	59.17	62.00
Rex (Krueger et al. 2020)*	74.31	66.20
Cumix (Mancini et al. 2020)*	76.78	74.74
DANN (Ganin et al. 2016)*	75.59	72.23
JiGen (Carlucci et al. 2019)*	84.95	79.45
CNBB (He, Shen, and Cui 2020)*	78.16	77.39
DecAug	***85.23***	***80.12***

* Implemented by ourselves.

- **Ablation study on PACS**

PACS	Art Painting	Cartoon	Sketch	Photo	Average
DecAug without orth loss	78.42	78.32	72.13	94.19	80.77
DecAug (orth 0.0005)	77.49	77.43	74.32	94.07	80.76
DecAug (orth 0.001)	78.12	77.34	76.97	94.91	81.83
DecAug (orth 0.01)	***79.00***	***79.61***	***75.64***	***95.33***	***82.39***

- **DecAug variants on PACS**

Model	Average
DecAug (DANN loss)	81.00
DecAug (orth between features)	79.90
DecAug (gradient-based orth)	***82.39***

- **The gradient visualization of the decomposed category-related and context-related features.**